

Google's SEED RL framework has been open sourced

Researchers at Google have open sourced the SEED RL framework. It can scale up artificial intelligence model training across thousands of machines.

Google researchers noted in a paper that this will enable AI algorithm training to be performed at millions of frames per second. It will also reduce the cost of doing so by as much as 80 per cent.

SEED RL is built atop the TensorFlow 2.0 framework. It leverages a combination of graphics processing units and tensor processing units to centralise model inference. After that, the inference is performed centrally using a learner component that trains the model.



The target model's variables and state information are kept local. The observations on them are sent to the learner at every step of the process. It also uses a network library based on the open source universal Remote Procedure Call (RPC) framework to minimise latency.

Google evaluated SEED RL's efficiency by benchmarking it on the Arcade Learning Environment, the Google Research Football environment, and various DeepMind Lab environments. As per results, it managed to solve a Google Research Football task while training the model at 2.4 million frames per second using 64 Cloud Tensor Processing Unit chips.

A hackathon to design an open source respirator to fight coronavirus

By way of its 'ultimate medical hackathon', Hackaday has called upon the open source community to support the creation of a blueprint for an open source respirator that could be manufactured and installed to treat those suffering from coronavirus. A respirator is a gadget that safeguards the wearer from breathing in hazardous air, including particulate matter, airborne micro-organisms, etc.



As observed in Italy, hospitals faced obstacles in maintaining the medical apparatus that provided respiratory assistance to patients. Suppliers have not been able to provide the hospital staff with valves because of the coronavirus outbreak.

A hospital in Brescia, a city in northern Italy where the coronavirus outbreak has been most severe, took the help of a local firm specialising in 3D printing to start using 3D printed valves for its 'reanimation' machines. Cristian Fracassi, chief executive of Isinnova, revealed that he reverse-engineered and 3D-printed the valves as the usual supply chain couldn't keep up with the hospital's unexpected huge demand. (*Business Insider* has reported that the patent-holders have threatened to sue Fracassi for copyright infringement.)

This hackathon from Hackaday is an open and urgent call to design a certain type of respirators. There are certain breathing apparatuses that aerosolize the virus, diffusing it into the air and hence harming those around the infected. To battle this, Hackaday has specified that Nasal cannula-based NIV systems are required. These humidify the air, mix it with oxygen and then push a constant stream of it into the lungs of the user.

If a simple working system can be designed, it will definitely give a major boost to manufacturers all over the world, positively impacting the fight against COVID-19.

GitHub launches its mobile app

The mobile app for developers and other team members working on projects in GitHub repositories is now available for users of iOS and Android.

Ryan Nystrom, director of engineering, had announced the application last November. Since then, it has been in the beta version.

Coding on a handheld device is generally not a pleasant experience, but this GitHub app has been aimed at all the other things that developers do, like raising or commenting on issues, approving pull requests, responding to notifications such as @mentions, etc.